Adaptive Heuristic Portfolio TSP Report

Overview

- Condition count: 8.
- Best held-out TSPLIB gap: phase8_best_single_fixed_heuristic.
- Best combined transfer gap: phase8_oracle_selector.
- Lowest selector regret: phase8_best_single_fixed_heuristic.
- Pareto-efficient conditions: phase8_full_solver_evolution, phase8_oracle_selector.

Run Metadata

- run_name: run_20260521_003344_t
- started_at_local: 2026-05-21 00:33:44
- finished_at_local: 2026-05-21 01:05:43
- duration_hhmm: 00:32
- duration_seconds: 1919.015
- seed_offset: 19000
- replicate_label: t
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Mode	Replay	Final TSPLIB Gap	Selector Regret	Runtime (ms)	Runtime-Adjusted Gap	Family Gap	Transfer Gap	Mean Novelty	Mean Complexity	Pareto Efficient
phase8_best_single_fixed_heuristic	best_fixed	none	0.07226	0.0	1486.861357	0.07226	0.007284	0.031222	0.0	0.0	False
phase8_random_portfolio	random_portfolio	none	0.187272	0.115012	3647.311786	0.459383	0.080217	0.119659	0.0	0.0	False
phase8_oracle_selector	oracle_selector	none	0.07226	0.0	1473.207443	0.07226	8.9e-05	0.026678	0.0	0.0	True

Condition	Mode	Replay	Final TSPLIB Gap	Selector Regret	Runtime (ms)	Runtime-Adjusted Gap	Family Gap	Transfer Gap	Mean Novelty	Mean Complexity	Pareto Efficient
phase8_supervised_ml_selector	supervised_selector	none	0.07226	0.0	1730.919671	0.084121	0.007284	0.031222	0.0	0.0	False
phase8_llm_static_selector	static_selector	none	0.214311	0.142051	485.304986	0.214311	0.048232	0.109419	0.0	2.36	False
phase8_llm_evolved_adaptive_controller	adaptive_controller	none	0.084733	0.012473	4204.717843	0.239618	0.001913	0.032426	0.450543	0.62	False
phase8_llm_evolved_controller_diversity_failure_replay	adaptive_controller	diversity_failure	0.21578	0.14352	588.777071	0.21578	0.129601	0.161351	0.623204	0.62	False
phase8_full_solver_evolution	full_solver	none	0.213387	0.141127	211.4116	0.213387	0.150269	0.173523	0.913819	0.56	True

Condition Notes

phase8_best_single_fixed_heuristic

- Execution mode best_fixed with replay none.
- Final held-out gap 0.07226, selector regret 0.0, runtime-adjusted gap 0.07226.
- Family transfer 0.007284 and combined transfer 0.031222.
- Runtime 1486.861357 ms, runtime inflation 0.0, mean novelty 0.0, and complexity 0.0.
- Pareto efficient: False.
- Fixed heuristic choice: farthest_insertion_two_opt.

phase8_random_portfolio

- Execution mode random_portfolio with replay none.
- Final held-out gap 0.187272, selector regret 0.115012, runtime-adjusted gap 0.459383.
- Family transfer 0.080217 and combined transfer 0.119659.
- Runtime 3647.311786 ms, runtime inflation 1.453027, mean novelty 0.0, and complexity 0.0.
- Pareto efficient: False.

phase8_oracle_selector

- Execution mode oracle_selector with replay none.
- Final held-out gap 0.07226, selector regret 0.0, runtime-adjusted gap 0.07226.

- Family transfer 8.9e-05 and combined transfer 0.026678.
- Runtime 1473.207443 ms, runtime inflation -0.009183, mean novelty 0.0, and complexity 0.0.
- Pareto efficient: True.

phase8_supervised_ml_selector

- Execution mode supervised_selector with replay none.
- Final held-out gap 0.07226, selector regret 0.0, runtime-adjusted gap 0.084121.
- Family transfer 0.007284 and combined transfer 0.031222.
- Runtime 1730.919671 ms, runtime inflation 0.164143, mean novelty 0.0, and complexity 0.0.
- Pareto efficient: False.

phase8_llm_static_selector

- Execution mode static_selector with replay none.
- Final held-out gap 0.214311, selector regret 0.142051, runtime-adjusted gap 0.214311.
- Family transfer 0.048232 and combined transfer 0.109419.
- Runtime 485.304986 ms, runtime inflation -0.673604, mean novelty 0.0, and complexity 2.36.
- Pareto efficient: False.
- Controller signature: nearest_neighbor_multistart:cluster_first_local_search:edge_preserving_restart:nearest_neighbor_multistart:stag3:cand2:restart1.
- Controller rule summary: {"acceptance_bias": 0.0, "candidate_limit_offset": 2, "clustered_heuristic": "cluster_first_local_search", "corridor_heuristic": "limited_three_opt", "default_heuristic": "farthest_insertion_two_opt", "description": "Deterministic one-shot hyper-heuristic selector over a frozen portfolio; instance-conditioned structure rules with bounded schedule tuning.", "failed_perturbation_threshold": 2, "fallback_rules": [{"then": "nearest_neighbor_multistart", "when": {"dimension_max": 70}}, {"then": "farthest_insertion_two_opt", "when": {"distance_cv_min": 0.55}}, {"then": "candidate_pruned_two_opt", "when": {"distance_cv_max": 0.35}}], "grid_like_heuristic": "candidate_pruned_two_opt", "instance_rules": [{"then": "edge_preserving_restart", "when": {"bottleneck_score_min": 0.6, "two_cluster_bottleneck_score_min": 0.58}}, {"then": "cluster_first_local_search", "when": {"cluster_separation_min": 0.4, "clustering_score_min": 0.45}}, {"then": "candidate_pruned_two_opt", "when": {"grid_likeness_min": 0.35}}, {"then": "limited_three_opt", "when": {"corridor_score_min": 0.45}}, {"then": "annealed_multi_start", "when": {"nearest_neighbor_trap_score_min": 0.25, "nn_distance_cv_max": 0.35}}, {"then": "annealed_multi_start", "when": {"nearest_neighbor_trap_score_min": 0.22}}], "low_improvement_threshold": 0.08, "name": "tsp_hh_oracle_portfolio_one_shot_deterministic_v1", "nearest_neighbor_trap_heuristic": "annealed_multi_start", "random_like_heuristic": "nearest_neighbor_multistart", "restart_offset": 1, "stagnation_switch_heuristic": "nearest_neighbor_multistart", "stagnation_threshold": 3, "temperature_scale": 1.1, "time_budget_trigger": 0.55, "two_cluster_bottleneck_heuristic": "edge_preserving_restart"}

phase8_llm_evolved_adaptive_controller

- Execution mode adaptive_controller with replay none.
- Final held-out gap 0.084733, selector regret 0.012473, runtime-adjusted gap 0.239618.
- Family transfer 0.001913 and combined transfer 0.032426.
- Runtime 4204.717843 ms, runtime inflation 1.827915, mean novelty 0.450543, and complexity 0.62.
- Pareto efficient: False.

- Controller signature: nearest_neighbor_multistart:cheapest_insertion_two_opt:farthest_insertion_two_opt:annealed_multi_start:stag3:cand-1:restart2.
- Controller rule summary: {"acceptance_bias": 0.008, "candidate_limit_offset": -1, "clustered_heuristic": "cheapest_insertion_two_opt", "corridor_heuristic": "limited_three_opt", "default_heuristic": "farthest_insertion_two_opt", "description": "Deterministic instance-adaptive scheduler over a frozen TSP heuristic portfolio; selects interpretable backbone based on structure descriptors and triggers a controlled diversification switch on stagnation/failure.", "failed_perturbation_threshold": 2, "grid_like_heuristic": "candidate_pruned_two_opt", "low_improvement_threshold": 0.08, "name": "interpretable_instance_adaptive_tsp_hh_v4", "nearest_neighbor_trap_heuristic": "farthest_insertion_two_opt", "random_like_heuristic": "nearest_neighbor_multistart", "restart_offset": 2, "stagnation_switch_heuristic": "annealed_multi_start", "stagnation_threshold": 3, "temperature_scale": 1.2, "time_budget_trigger": 0.55, "two_cluster_bottleneck_heuristic": "farthest_insertion_two_opt"}

phase8_llm_evolved_controller_diversity_failure_replay

- Execution mode adaptive_controller with replay diversity_failure.
- Final held-out gap 0.21578, selector regret 0.14352, runtime-adjusted gap 0.21578.
- Family transfer 0.129601 and combined transfer 0.161351.
- Runtime 588.777071 ms, runtime inflation -0.604013, mean novelty 0.623204, and complexity 0.62.
- Pareto efficient: False.
- Controller signature: annealed_multi_start:cluster_first_local_search:edge_preserving_restart:annealed_multi_start:stag3:cand-3:restart2.
- Controller rule summary: {"acceptance_bias": 0.01, "candidate_limit_offset": -3, "clustered_heuristic": "cluster_first_local_search", "corridor_heuristic": "limited_three_opt", "default_heuristic": "farthest_insertion_two_opt", "description": "Deterministic instance-adaptive scheduler/tuner over frozen TSP heuristic portfolio with interpretable switches based on structure scores and online stagnation (diversity_failure replay-hardened).", "failed_perturbation_threshold": 3, "grid_like_heuristic": "candidate_pruned_two_opt", "low_improvement_threshold": 0.1, "name": "interpretable_portfolio_controller_v2", "nearest_neighbor_trap_heuristic": "nearest_neighbor_multistart", "random_like_heuristic": "annealed_multi_start", "restart_offset": 2, "stagnation_switch_heuristic": "annealed_multi_start", "stagnation_threshold": 3, "temperature_scale": 1.2, "time_budget_trigger": 0.6, "two_cluster_bottleneck_heuristic": "edge_preserving_restart"}

phase8_full_solver_evolution

- Execution mode full_solver with replay none.
- Final held-out gap 0.213387, selector regret 0.141127, runtime-adjusted gap 0.213387.
- Family transfer 0.150269 and combined transfer 0.173523.
- Runtime 211.4116 ms, runtime inflation -0.857814, mean novelty 0.913819, and complexity 0.56.
- Pareto efficient: True.

Judge Appendix

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Conservative Interpretation of TSP Adaptive Heuristic Portfolio Results

Primary Endpoint: Held-out TSPLIB Gap

- The **best single fixed heuristic** (farthest_insertion_two_opt) achieves a TSPLIB gap of **0.07226**, matching the **oracle selector** performance exactly.

- The **supervised ML selector** also matches this TSPLIB gap (0.07226), but with slightly higher runtime (1731 ms vs. ~1473 ms oracle).
- The **LLM-evolved adaptive controller** shows a slightly higher gap (0.08473) with significant runtime inflation (~4205 ms) compared to oracle and fixed heuristic.
- Other adaptive or static selectors have notably worse TSPLIB gaps (≥0.21), with lower or comparable runtimes.

Selector Regret vs. Oracle Portfolio

- **Oracle selector** has zero regret (by definition).
- **Best single fixed heuristic** also has zero measured selector regret—indicating no regret since it is a fixed heuristic.
- **Supervised ML selector** has zero regret, suggesting it effectively approaches oracle choice.
- **LLM adaptive controllers** show small positive regret (0.0125 for evolved adaptive, ~0.14 for others), indicating imperfect adaptation.

Runtime-Adjusted Gap and Runtime Inflation

- Oracle and best fixed heuristic yield the minimal runtime-adjusted gap (0.07226) with near-zero or slight negative runtime inflation.
- Supervised ML selector incurs moderate runtime inflation (+16%) with a slightly higher runtime-adjusted gap (0.0841).
- The best LLM adaptive controller has a higher runtime-adjusted gap (0.24) and substantial runtime inflation (+183%)—runtime inflation is not negligible.
- Other adaptive or full solver controllers have runtime-adjusted gaps around 0.21 but with negative runtime inflation (faster). However, their actual TSPLIB gaps are much larger, indicating worse solution quality despite faster runtimes.

Pareto Efficiency

- Only the **oracle selector** and **full solver evolution** are marked Pareto efficient.
- The full solver evolution has worse TSPLIB gap but minimal runtime and low complexity.
- Oracle selector achieves best TSPLIB gap with moderate runtime.
- Best fixed heuristic is not Pareto efficient here, but closely matches oracle gap with minimal runtime.

Cross-Family Transfer

- Transfer gaps (performance loss on out-of-family instances) are lowest for oracle (~0.027) and best fixed heuristic (~0.031), indicating good generalization.
- The best adaptive controller transfer gap is small (~0.032), but worse controllers or full solver have substantially larger transfer gaps (~0.16–0.17+), reflecting poorer generalization.

Summary & Recommendations

- The **best single fixed heuristic (farthest_insertion_two_opt) already achieves held-out TSPLIB gaps matching the oracle selector**, suggesting limited room for improvement from adaptation in terms of final solution quality.
- The **oracle selector serves as a strong upper bound** but is not deployable; fixed heuristic or supervised ML selectors closely approximate it with modest runtime costs.

- **Adaptive controllers studied show interpretable instance-adaptive control with some success but with increased runtime and selector regret**, without surpassing the fixed heuristic baseline on held-out TSPLIB.
- **Runtime inflation is a critical trade-off; adaptive controllers often incur substantial runtime increases without commensurate quality improvement**, limiting their practical advantage.
- **Static and ML selectors that approach oracle performance without excessive runtime inflation are preferable for deployment.**
- Claims of new algorithmic discovery should be avoided—results support **effective hyper-heuristic controller design over known heuristics**, highlighting interpretability and controlled adaptation rather than breakthrough heuristics.
- Cross-family transfer performance aligns with TSPLIB gap trends, reinforcing robustness of fixed and oracle selectors.

Key Data Points:

Condition	TSPLIB Gap	Runtime ms	Runtime-Adjusted Gap	Selector Regret	Runtime Inflation	Pareto Efficient	Transfer Gap
phase8_best_single_fixed_heuristic	0.07226	1487	0.07226	0.0	0.0	No	0.03122
phase8_oracle_selector	0.07226	1473	0.07226	0.0	-0.009	Yes	0.02668
phase8_supervised_ml_selector	0.07226	1731	0.08412	0.0	+0.164	No	0.03122
phase8_llm_evolved_adaptive_controller	0.08473	4205	0.23962	0.01247	+1.828	No	0.03243
phase8_full_solver_evolution	0.21339	211	0.21339	0.14113	-0.858	Yes	0.17352

Conclusion

- For deployable hyper-heuristic control of TSP portfolios, **the best fixed heuristic or supervised ML selectors best balance interpretability, near-oracle quality, and runtime efficiency.**
- **LLM-based adaptive controllers demonstrate interpretable adaptation but suffer from excessive runtime and modest quality gains, limiting practical appeal at this stage.**
- Further improvement should focus on reducing runtime inflation or improving selector regret without sacrificing interpretability.